

**Technical Description of the Humoral Immune Response Following
Rabies Vaccination and Methodological Basis of Virus Neutralization
Tests (RFFIT and FAVN) in Companion Animals**

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Abstract

Objective. To describe the humoral immune response mechanism following rabies vaccination and the methodological basis of RFFIT and FAVN virus-neutralizing antibody tests in small animals. **Method.** Review of rabies vaccination immunology, B-lymphocyte activation, specific IgG production, and the standardized protocols of both serological techniques. **Results.** RFFIT and FAVN are functionally equivalent but differ in biosafety requirements, accessibility and cost: RFFIT requires live virus at BSL-2/3; FAVN uses a fixed strain under more controlled conditions. Both quantify rVNA by cytopathic effect reduction. The protective threshold of 0.5 IU/mL applies to both tests. **Conclusion.** The choice between RFFIT and FAVN depends on the accredited laboratory available in the country of origin; clinical and regulatory interpretation is identical for both.

Keywords: humoral immune response, RFFIT, FAVN, rabies IgG, rVNA, 0.5 IU/mL, BSL-2, accredited laboratory

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Descriptive technical review -- Zoovet Travel Technical Series, Volume II

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Keywords: rabies neutralizing antibody titre; RFFIT; FAVN; humoral immune response rabies vaccination; seroconversion dogs; 0.5 IU/mL threshold; primary vaccine failure; movement of companion animals; rabies serology methodology; inter-laboratory variability

1. Introduction

Rabies vaccination of companion animals constitutes the primary biosecurity barrier against introduction of the rabies virus (RABV) into disease-free territories through international pet movement. For this vaccination to fulfil its regulatory function -- enabling legally valid certification as well as animal identification (microchip) under frameworks such as EU Regulation (EU) 576/2013, the CDC/USDA rules effective August 2024, and equivalent instruments in the UK, Australia, and Japan -- the immune response it generates must be verifiable by an objective, reproducible, internationally standardized laboratory method.

That method is the serum virus neutralization test (VNT), operationalized in two principal forms: the Rapid Fluorescent Focus Inhibition Test (RFFIT) and the Fluorescent Antibody Virus Neutralization test (FAVN). Both measure the functional capacity of serum antibodies to inhibit RABV infection of cell cultures in vitro, and both report results in International Units per millilitre (IU/mL) standardized against a WHO reference serum.

The threshold of ≥ 0.5 IU/mL, universally adopted by WOAH, WHO, and the major national import frameworks, is the regulatory decision criterion that governs whether an animal's serology result is considered adequate for international movement purposes.

The present descriptive review is the second in the Zoovet Travel Technical Series. Where Volume I addressed the regulatory and immunological basis of the 30-day post-primary vaccination sampling interval, this document focuses on the underlying immunological sequence that produces the antibodies these tests measure, the methodological principles of the assays themselves, and the documented sources of variability that practitioners must understand to correctly interpret results and anticipate potential failure scenarios.

2. Humoral Immune Response Following Primary Rabies Vaccination

2.1. Primary Immune Activation: From Antigen to Lymphocyte

Commercially available veterinary rabies vaccines are, with very rare exceptions, formulated as inactivated whole-virus or subunit preparations with adjuvant (WOAH, 2024; Tizard, 2021). Following intramuscular or subcutaneous administration, the vaccine antigen -- principally the RABV glycoprotein (G-protein), which carries the major neutralization epitopes -- is taken up by professional antigen-presenting cells (APCs), primarily dendritic cells and macrophages, at the injection site and in regional lymph nodes (Flamand et al., 1993; Day, 2007).

The processed antigen is presented to naive CD4⁺ T helper lymphocytes via major histocompatibility complex class II (MHC-II) molecules. This interaction, requiring co-stimulatory signals, initiates clonal expansion of antigen-specific T helper cells. Simultaneously, RABV antigen may directly activate specific B lymphocytes through B-cell receptor engagement. The T helper-B cell interaction in germinal centres of secondary lymphoid organs initiates the primary antibody response (Day, 2007; Tizard, 2021).

The literature consistently describes this initial phase -- from antigen administration to first detectable serum antibody -- as a lag phase during which immunological processing occurs but circulating antibody is not yet measurable at the ≥ 0.5 IU/mL threshold in most animals. The duration of this phase is not fixed: it is influenced by adjuvant type, antigen dose, route of administration, and host immunological competence (Day, 2007; Kennedy et al., 2007).

2.2. IgM Production, Class Switching, and IgG Consolidation

The initial antibody class produced in the primary immune response is IgM, a pentameric molecule capable of efficient viral agglutination but with relatively lower affinity compared to subsequently produced IgG isotypes (Tizard, 2021). IgM antibodies are measurable earlier in the response and contribute to initial virus neutralization. The VNTs used in regulatory certification (RFFIT and FAVN) detect neutralizing antibodies regardless of isotype; however, IgM-dominated responses may produce titres below the regulatory threshold even in animals that will subsequently generate a robust IgG response.

As the primary response matures, activated B lymphocytes in germinal centres undergo class switch recombination -- a molecular process by which the antibody constant region gene is replaced, shifting production from IgM to IgG (and, in some B cell lineages, IgA or IgE). Concurrently, affinity maturation -- the somatic hypermutation and selection process operating in germinal centres -- progressively increases the binding affinity of IgG antibodies for RABV epitopes (Tizard, 2021; Day, 2007). The result is a population of long-lived plasma cells secreting high-affinity anti-RABV IgG, which constitutes the dominant antibody class measured by RFFIT and FAVN at the regulatory sampling timepoint of ≥ 30 days post-primary vaccination.

2.3. Temporal Kinetics of Post-Vaccination Antibody Development

The scientific literature documents post-vaccination antibody kinetics in dogs using both experimental challenge models and large-scale field seroprevalence studies, from which the following general description is synthesised. It is important to note that the published

evidence reflects population-level trends and substantial individual variability; the intervals described below are ranges observed across studies, not fixed physiological constants.

Wallace et al. (2017), in the largest published analysis of primary rabies vaccination outcomes in dogs ($n = 8,011$), documented that failure to reach ≥ 0.5 IU/mL was associated with shorter intervals between vaccination and sampling, among other factors. The study demonstrated that the relationship between time post-vaccination and serological adequacy is not linear and is heavily modulated by age, body size, and the number of prior vaccinations.

It is pertinent to note that the published literature does not support a single "peak day" for primary vaccination antibody response that is universal across all dogs, all vaccines, and all conditions. Various studies have observed initial seroconversion in some proportion of dogs within the first week, while others document that a meaningful fraction of animals do not reach ≥ 0.5 IU/mL until the third or fourth week post-vaccination. This biological heterogeneity is precisely the rationale that underlies the regulatory requirement for a ≥ 30 -day minimum sampling interval, as described in Volume I of this series.

2.4. Immunological Memory Following Primary Vaccination

A central immunological outcome of successful primary vaccination is the generation of immunological memory: populations of long-lived memory B lymphocytes and memory CD4⁺ and CD8⁺ T lymphocytes that persist in secondary lymphoid organs and peripheral circulation for months to years after vaccination (Day, 2007; Tizard, 2021). Upon re-exposure to RABV antigen -- whether through a booster vaccination or natural exposure -- these memory cells mount a qualitatively and quantitatively superior secondary immune response: shorter lag phase, more rapid antibody production, higher peak titres, and greater affinity of IgG antibodies.

This distinction between primary and secondary responses has direct operational significance for international pet movement: animals receiving a valid booster vaccination within the period of coverage of a previous dose do not require a 30-day post-vaccination

waiting period before regulatory serology sampling, as their pre-existing memory ensures rapid and robust antibody restoration. Regulatory frameworks including EU Reg. 576/2013 and CDC rules explicitly distinguish these two scenarios.

3. The ≥ 0.5 IU/mL Threshold: Technical Basis and Interpretive Limitations

3.1. Origin and Scientific Foundation

The threshold of ≥ 0.5 International Units per millilitre as the regulatory criterion for adequate rabies immunity was not derived from a single landmark study but emerged from the progressive accumulation of challenge experiment data, vaccine efficacy evaluations, and seroneutralization method standardization work conducted through the latter decades of the twentieth century (Moore & Hanlon, 2010; Briggs et al., 1998).

Moore & Hanlon (2010), in a comprehensive analysis of rabies-specific antibodies as surrogates of protection, reviewed the evidence base underlying the 0.5 IU/mL criterion and concluded that, at the population level, animals with serum titres at or above this threshold demonstrate substantially reduced risk of developing clinical rabies following challenge. The threshold was adopted by the WHO and the Office International des Épizooties (now WOAHA) and formalized in the WOAHA Terrestrial Animal Health Code as the internationally recognized correlate of adequate rabies vaccination response.

A critical interpretive nuance documented in the peer-reviewed literature is that ≥ 0.5 IU/mL constitutes an operational correlate of protection at the population level -- not a binary guarantee of sterile immunity in every individual animal above the threshold (Moore & Hanlon, 2010). Animals below the threshold are not necessarily unprotected (cellular immunity may contribute to resistance), and animals above the threshold are not absolutely guaranteed to survive every potential exposure scenario. The threshold is a regulatory decision criterion, selected for its practical utility and reproducibility, not as a biological certainty marker.

3.2. The IU/mL Unit: Calibration and Standardization

Results of RFFIT and FAVN are expressed in International Units per millilitre (IU/mL), standardized against a WHO/WOAH reference serum of defined titre (currently the WHO reference serum 2nd standard, 30 IU/mL). This standardization -- described in the WOAH Manual of Diagnostic Tests and Vaccines for Terrestrial Animals (2024) and operationalized through the ISO 17025-accredited approved laboratory system -- enables results from different laboratories and different geographic locations to be compared on a common scale. The annual proficiency testing programme coordinated by ANSES-EURL (Wasniewski et al., 2019) is the principal mechanism by which inter-laboratory comparability of the ≥ 0.5 IU/mL decision point is verified and maintained internationally.

4. Methodological Basis of the Rapid Fluorescent Focus Inhibition Test (RFFIT)

4.1. Principle and Technical Protocol

The Rapid Fluorescent Focus Inhibition Test was developed by Smith, Yager, and Baer (1973) as a microtest adaptation of the original mouse neutralization test, replacing in vivo mouse inoculation with in vitro cell culture methodology. RFFIT is the method of record for the WHO and ACIP for human rabies serology and is widely used in approved laboratories for companion animal certification (Moore & Hanlon, 2010; Moore, 2018).

The principle of RFFIT is based on the capacity of antibodies in test serum to neutralize a fixed dose of live RABV challenge virus standard (CVS-11 or equivalent) and thus prevent infection of a susceptible cell monolayer (typically mouse neuroblastoma cells, MNA). The procedure involves:

- Serial dilution of test serum and reference standard in a microplate.
- Addition of a fixed titre of CVS challenge virus to each well.
- Incubation to allow antibody-virus binding (neutralization).
- Addition of susceptible cells and further incubation.

- Fixation and staining with fluorescein isothiocyanate (FITC)-labelled anti-RABV nucleoprotein antibody.
- Fluorescence microscopy to count infected foci (fluorescent cells).
- Calculation of antibody titre (IU/mL) against the reference serum using the Reed-Muench or Kärber method.

Because RFFIT requires the use of live RABV, it must be performed in a laboratory with appropriate biosafety infrastructure (BSL-2 minimum, BSL-3 in many jurisdictions) by personnel trained in the technique. The requirement for infectious virus and specialized equipment limits RFFIT to designated reference and approved laboratories (Moore, 2018; Wasniewski et al., 2019).

4.2. Analytical Considerations and Known Limitations

Technical parameters that affect RFFIT performance include: the batch and titre of CVS challenge virus; the concentration and passage number of the cell line; incubation temperature and CO₂ concentration; the choice of FITC conjugate; and the visual or automated fluorescence reading method (Moore, 2018). Standardization of these variables across laboratories is one of the primary aims of the ANSES-EURL annual proficiency testing programme (Wasniewski et al., 2019). Briggs et al. (1998) documented that test-to-test variation (intra-laboratory) is most pronounced for sera with titres in the low-to-moderate range, specifically for RFFIT titres below 1.0 IU/mL -- a range that includes the regulatory 0.5 IU/mL threshold and makes repeat testing of borderline samples a prudent quality practice.

5. Methodological Basis of the Fluorescent Antibody Virus Neutralization Test (FAVN)

5.1. Development and Principle

The Fluorescent Antibody Virus Neutralization test was developed and described by Cliquet, Aubert, and Sagné (1998) at CNEVA Nancy (now ANSES Nancy Laboratory for Rabies and Wildlife -- the EURL). The FAVN test is a microadaptation of the RFFIT, designed for higher-throughput processing, and has become the predominant method in European approved laboratories for companion animal export serology.

The technical principle of FAVN is functionally equivalent to RFFIT: serial dilution of test serum; addition of a fixed dose of CVS challenge virus; cell culture incubation; fixation and FITC staining; fluorescence microscopy; Reed-Muench titre calculation (Cliquet et al., 1998). Key technical differences include plate format (96-well microplates optimized for FAVN), cell type (often BHK-21 cells in the original FAVN protocol versus MNA cells in the standard RFFIT), and minor incubation parameters (Cliquet et al., 1998; Wasniewski et al., 2019).

5.2. Recognized Equivalence with RFFIT

The original development paper (Cliquet et al., 1998) demonstrated 100% specificity using 414 sera from rabies-free areas and satisfactory accuracy against sera of known titres, with concordant results among FAVN, RFFIT, and the mouse neutralization test for comparative serum panels. The WOAHA Terrestrial Manual (2024) formally recognizes FAVN and RFFIT as producing equivalent results for regulatory certification purposes, which underlies their interchangeability in approved laboratory networks worldwide.

Briggs et al. (1998) explicitly confirmed this equivalence in a comparative study of 168 unvaccinated and 70 high-titre vaccinated dog and cat sera: no significant difference was observed in sensitivity or specificity between RFFIT and FAVN for unvaccinated animals or high-titre vaccinated animals. For the subset of 95 sera with low-to-moderate RFFIT titres (< 1.0 IU/mL), test-to-test variation was documented for both methods equally, with no statistically significant direction of discordance favouring either assay.

■ Key comparative finding -- RFFIT vs FAVN Table 2. RFFIT vs FAVN: principal technical characteristics and recognized equivalence

Parameter	RFFIT	FAVN	Origin
Smith,			

Yager & Baer (1973) -- adapted from mouse neutralization test
 Cliquet, Aubert & Sagné (1998) -- microadaptation of RFFIT Principle
 Serum antibody neutralization of CVS; FITC staining of infected foci
 Identical principle; different plate format and cell line parameters
 Virus CVS-11 or equivalent (live RABV required) CVS-11 or equivalent (live RABV required)
 Cell line Typically MNA (mouse neuroblastoma) Typically BHK-21 in original protocol; adaptations exist
 Result unit IU/mL vs WHO reference serum IU/mL vs WHO reference serum
 Regulatory acceptance WHO, WOA, CDC/USDA, EU, UK, AU, JP
 WHO, WOA, CDC/USDA, EU, UK, AU, JP Recognized equivalence Yes -- WOA Manual (2024); Briggs et al. (1998); no statistically significant difference in sensitivity/specificity at ≥ 0.5 IU/mL decision point
 Known variability zone Low-to-moderate titre range (< 1.0 IU/mL RFFIT); test-to-test variation documented for both methods equally (Briggs et al., 1998)
 Biosafety requirement BSL-2/3 (live virus) BSL-2/3 (live virus)
 Inter-lab quality Annual proficiency testing by ANSES-EURL (ISO/IEC 17025; ISO 13528) -- mandatory for EU-approved laboratories (Wasniewski et al., 2019)
 Abbreviations: CVS = Challenge Virus Standard; FITC = Fluorescein Isothiocyanate; MNA = Mouse Neuroblastoma; BHK = Baby Hamster Kidney; EURL = European Union Reference Laboratory. Source: Cliquet et al. (1998); Briggs et al. (1998); WOA Manual (2024); Wasniewski et al. (2019).

5.3. Inter-Laboratory Variability and Quality Assurance

Inter-laboratory variability in rVNA quantification is a documented and actively managed challenge in rabies serology networks. Wasniewski et al. (2019), reporting on the annual proficiency testing programme of the ANSES-EURL (the EU-designated coordinating laboratory for approved rabies serology laboratories under Commission Decision 2000/258/EC), described a PT design compliant with ISO 13528 and ISO/IEC 17043 international standards. The programme distributes standardized serum panels to participating approved laboratories annually, and results are evaluated against assigned values from compiled historical PT data. This ongoing quality framework is the primary

mechanism ensuring that an IU/mL result from a laboratory in Peru, Europe, or Asia is comparable and legally interchangeable for regulatory certification purposes.

6. Factors Modulating the Post-Vaccination Antibody Response

The scientific literature identifies multiple host- and product-related variables that influence the magnitude and timing of the post-vaccination rVNA response in dogs. These factors explain why primary vaccination outcomes are not uniform across animals and why population-level serology data always reflect a distribution -- not a single outcome -- of antibody responses.

Age at Vaccination Animals vaccinated before 16 weeks of age may show lower titres due to residual maternal antibody interference and relative immunological immaturity. Wallace et al. (2017) identified age < 1 year as a significant risk factor for inadequate antibody response to primary vaccination. Kennedy et al. (2007) confirmed age-related effects in a dataset of over 10,000 dogs.

Body Size and Breed The literature consistently documents higher failure rates for larger-breed dogs compared to smaller breeds (Kennedy et al., 2007; Berndtsson et al., 2011). The mechanism is not fully elucidated but likely reflects differences in antigen distribution and immune system scaling relative to body mass. Breed-specific immune response variation has also been reported.

Vaccine Formulation Not all inactivated rabies vaccines demonstrate equivalent immunogenicity in dogs under all conditions. Adjuvant type, antigen mass, and manufacturer-specific formulation differences can influence the speed and magnitude of the primary immune response. Kennedy et al. (2007) documented significant differences across vaccine products in the same large cohort.

Maternal Antibody Interference Passively acquired maternal antibodies against RABV can suppress the active primary immune response in young puppies. Published literature indicates that maternal antibody levels wane by 12-16 weeks of age in most pups, which explains why the minimum vaccination age in major regulatory frameworks is 12 weeks (EU Reg. 576/2013; WOA, 2023).

Nutritional and Health Status Concurrent infectious disease, parasitism, malnutrition, or

systemic illness at the time of vaccination impairs the immune response. Animals in poor general health have been observed to produce lower antibody responses following vaccination. These are individual clinical factors that a supervising veterinarian must assess prior to vaccination for export purposes. **Immunosuppressive Conditions and Drugs** Animals receiving corticosteroids, chemotherapy, or other immunosuppressive treatments may fail to mount an adequate primary response. Concurrent immunosuppressive disease (e.g., hyperadrenocorticism, leishmaniasis) can similarly attenuate the vaccine-induced immune response. Clinical screening before vaccination is essential in such cases. **Cold Chain Integrity** Inactivated rabies vaccines must be maintained at 2-8°C throughout storage and transport. Exposure to temperatures outside the recommended range, freezing, or prolonged heat exposure can result in partial or complete loss of antigen immunogenicity. The February 2026 recall of IMRAB® 3TF serial #18665 -- attributable to vials containing sterile water instead of vaccine -- illustrates an extreme but instructive case of product integrity failure at the manufacturing level (Boehringer Ingelheim, 2026). Routine cold-chain verification at the point of administration is a standard quality practice. **Interval Between Vaccination and Sampling** As described in Volume I of this series, the interval between vaccination and blood draw is a critical determinant of whether the measured titre reflects the consolidated antibody response or an early, potentially incomplete phase. Sampling within the first 21 days post-primary vaccination yields results that may not reflect the final plateau titre, with higher variability and failure rates at very early intervals (Wallace et al., 2017; Crozet et al., 2024).

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Boehringer Ingelheim initiated a voluntary recall of IMRAB® 3TF, 1 mL, Serial #18665 in early 2026 after a limited number of vials were found to contain sterile water instead of vaccine. The affected lot was distributed to veterinary clinics between September 2025 and January 2026 in the United States. Boehringer recommended immediate revaccination for any animal that received vaccine from this serial. This event is cited here not as a characterization of the product or manufacturer -- recall was voluntary and rapid -- but as a real-world illustration of the cold chain and product integrity principles discussed above. Animals from this lot would be expected to have no serological response and would require full protocol restart from vaccination. For practitioners managing international export, verification of vaccine batch documentation is therefore a non-trivial step (Boehringer Ingelheim, 2026; Massachusetts Veterinary Medical Association, 2026).

7. Primary Vaccine Failure and Non-Responders

7.1. Definition and Documented Existence

The veterinary immunology literature recognizes two principal forms of vaccination failure:

Primary vaccine failure refers to the inability of an animal to generate a detectable and adequate antibody response following correctly administered vaccination. This contrasts with secondary vaccine failure (waning immunity following a previously

adequate response) and apparent vaccine failure (failure attributable to incorrect vaccine storage, administration error, or sampling at an immunologically inappropriate timepoint).

Wallace et al. (2017) explicitly noted that published studies across different populations and contexts have reported that approximately 10% of immunologically naive dogs may not achieve ≥ 0.5 IU/mL following a single dose of rabies vaccine, with the proportion varying substantially depending on age, breed, vaccine product, and the interval between vaccination and sampling. This figure should not be interpreted as a universal constant: the literature reflects genuine heterogeneity, and reported rates depend critically on study design, population characteristics, and the timepoint at which serology was performed.

7.2. Factors Associated with Primary Vaccine Failure

Kennedy et al. (2007), in a dataset of over 10,483 dogs, identified statistically significant associations between primary vaccination failure and the following factors: age below one year; large body size; specific vaccine products (with significant variation across brands in the same cohort); and the number of doses received. Berndtsson et al. (2011) confirmed in a prospective Swedish cohort ($n = 6,789$) that two-dose vaccination protocols significantly reduced the proportion of dogs failing to reach ≥ 0.5 IU/mL compared with single-dose protocols, particularly in large breeds.

These findings have practical implications for dogs undergoing serology for international export: animals with risk factors for primary vaccine failure (young, large breed, single-dose protocol) may benefit from earlier identification of potential failure through appropriate scheduling, and from anticipating the possibility of a repeated vaccination cycle if the initial result is inadequate.

7.3. Recommended Technical Approach for Titres < 0.5 IU/mL

When a regulatory serology result falls below ≥ 0.5 IU/mL, the published consensus and the major regulatory frameworks (EU, CDC, WOA) indicate the following sequence: the animal requires revaccination (with a booster dose), followed by a new sampling event.

Under EU rules (Reg. 576/2013), a failed titre result requires a full restart of the 30-day post-vaccination sampling interval plus the 3-month pre-movement waiting period from the new sampling date. Under CDC rules, revaccination and a new sample drawn ≥ 30 days after the new vaccination are required before the animal can be eligible for importation without a 28-day quarantine.

A titre below ≥ 0.5 IU/mL does not necessarily mean the animal has no immunity. Cellular immunity and low-level antibody-mediated protection may still be present. However, for the purposes of regulatory certification in international pet movement, the VNT result of ≥ 0.5 IU/mL at an approved laboratory is the only accepted proof of adequate immune status. Clinical interpretation of VNT results must remain within the regulated framework for export purposes.

8. Limitations of This Document

- This is a descriptive review without meta-analytic methodology. Citation selection was purposive, not systematic, with the goal of representing the best-supported evidence base for the described concepts.
- The immunological kinetics described in Section 2 are derived from studies conducted predominantly in European and North American dog populations under specific study conditions. Generalizability to all geographic populations, tropical disease environments, and local breed compositions (including those prevalent in Peru) should be applied with appropriate caution.
- Regulatory texts cited in this document reflect the state of applicable rules as of early 2026. International pet movement regulations are subject to update; practitioners must verify current requirements with the relevant competent authority before initiating export protocols.
- This document does not address rabies serology in cats or ferrets in detail; while the immunological principles are broadly similar, species-specific considerations apply and readers should consult species-specific literature and relevant SmPCs.

- No clinical recommendation is made at the individual animal level. All clinical decisions must be made by a licensed veterinarian with access to the specific animal's history, health status, and applicable regulatory requirements.

9. Declaration of Scope

Scope and Limitations Statement This document is a descriptive technical review based on peer-reviewed scientific literature and publicly available regulatory texts. It presents no original experimental data, claims no new discoveries, and makes no individually-tailored clinical recommendations. All cited claims are traceable to the references listed below. The document does not substitute for specific veterinary evaluation, does not establish or modify any regulatory requirement, and should not be cited as primary regulatory guidance. Its purpose is to support the scientific literacy of veterinary practitioners engaged in international pet export consultation.

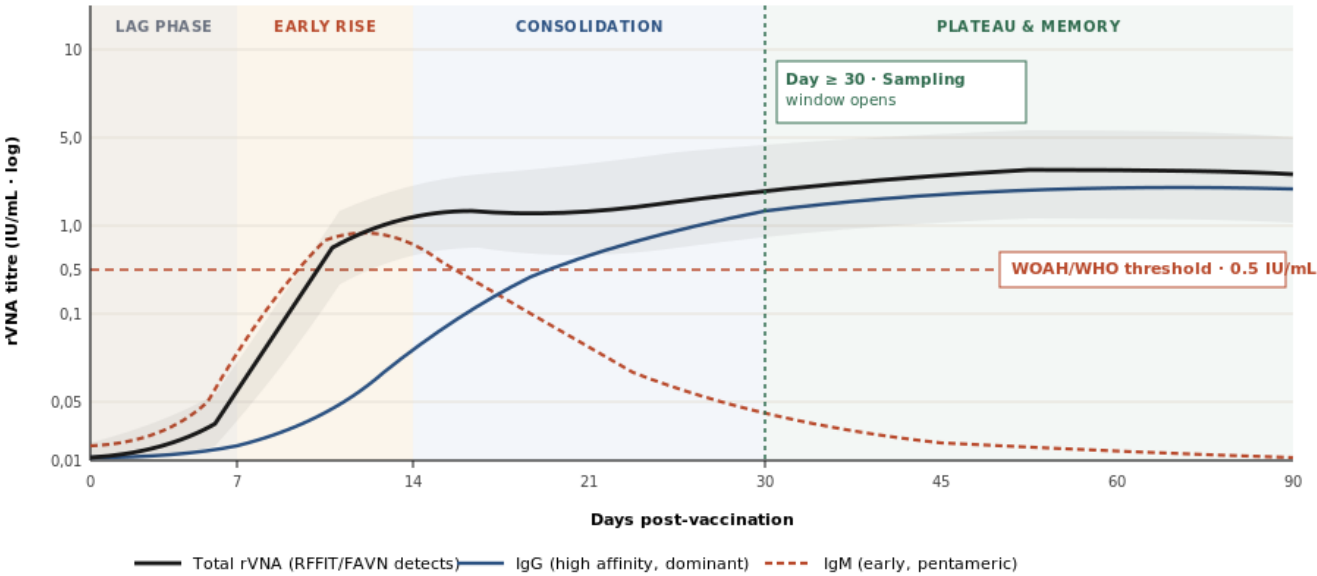
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rVNA neutralizing antibody kinetics · Primary rabies vaccination

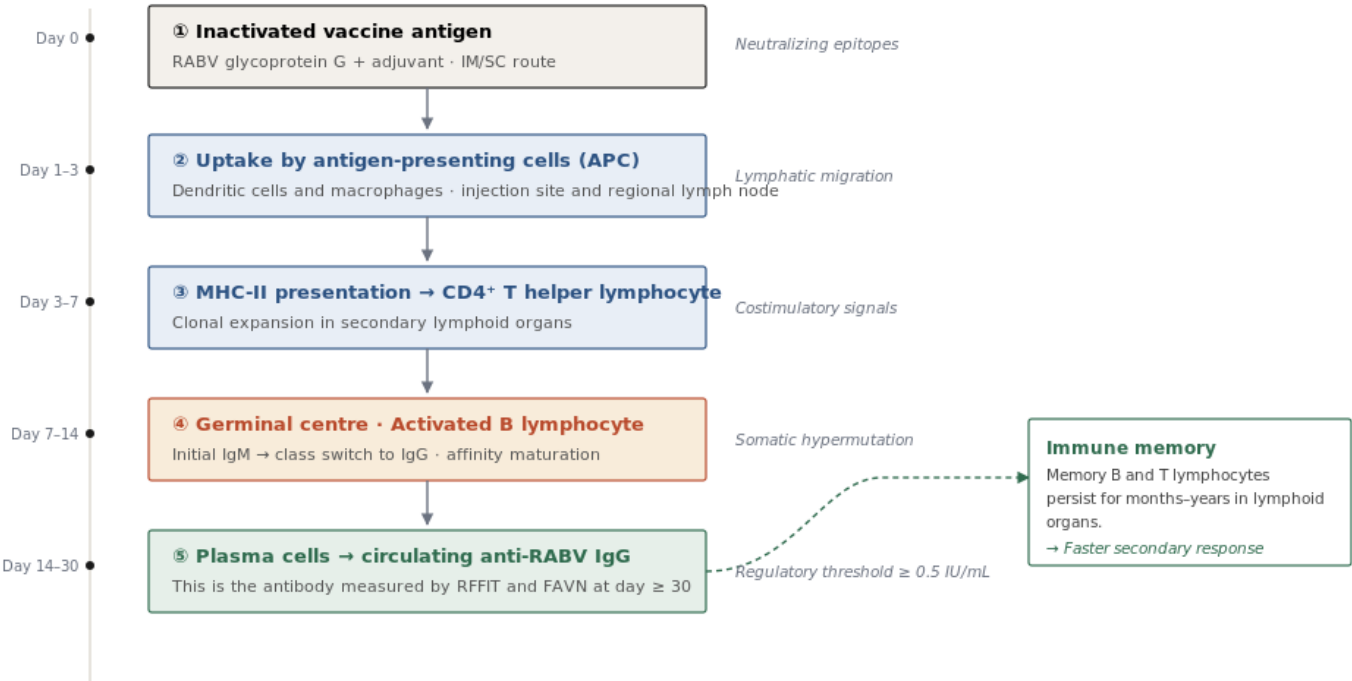
Descriptive synthesis of published literature — values are representative ranges, not individual constants



Sources: Wallace et al. 2017 (n=8,011) · Kennedy et al. 2007 · Crozet et al. 2024 · Berndtsson et al. 2011 · WOAH Terrestrial Manual 2024
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Primary humoral response cascade to rabies vaccine

From vaccine antigen to IgG anti-RABV plasma cells



Sources: Tizard 2021 · Day 2007 · Flamand et al. 1993 · Kennedy et al. 2007 · WOAHP Terrestrial Manual 2024
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RFFIT vs FAVN · WOAAH-recognized methodological equivalence

Both tests measure neutralizing antibodies (rVNA) in IU/mL against the WHO reference serum

RFFIT

Rapid Fluorescent Focus Inhibition Test · Smith, Yager & Baer (1973)

- 1

Serial dilution of serum
Microplate + WHO reference serum
- 2

Challenge virus addition
Fixed dose of CVS-11 (live RABV)
- 3

Antibody-virus incubation
Neutralization phase
- 4

Cell line: MNA (neuroblastoma)
Murine neuroblastoma, original cell type
- 5

Fixation + FITC staining
Anti-nucleoprotein FITC-conjugated
- 6

Fluorescence microscopy reading
Infected focus count
- 7

Titre calculation (Reed-Muench)
Result in IU/mL · threshold ≥ 0.5

FAVN

Fluorescent Antibody Virus Neutralization · Cliquet, Aubert & Sagné (1998)

- 1

Serial dilution of serum
96-well microplate · high throughput
- 2

Challenge virus addition
Fixed dose of CVS-11 (live RABV)
- 3

Antibody-virus incubation
Neutralization phase
- 4

Cell line: BHK-21 (hamster kidney)
CNEVA Nancy protocol — adaptations exist
- 5

Fixation + FITC staining
Anti-nucleoprotein FITC-conjugated
- 6

Fluorescence microscopy reading
Infected focus count
- 7

Titre calculation (Reed-Muench)
Result in IU/mL · threshold ≥ 0.5

RECOGNIZED REGULATORY EQUIVALENCE

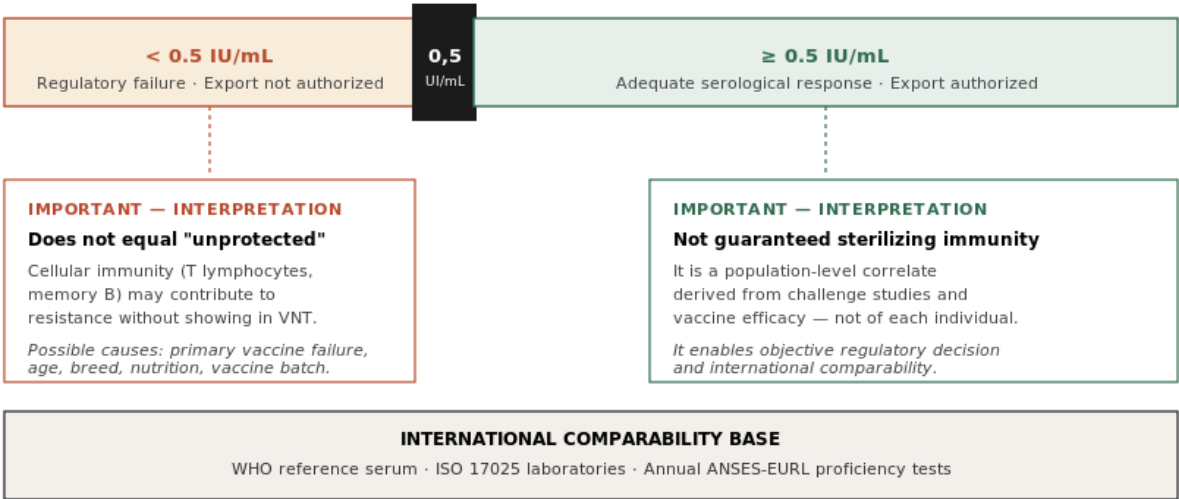
WOAH · WHO · EU Reg. 576/2013 · CDC/USDA · DAFF (AU) · MPI (NZ) · MAFF (JP)

Briggs et al. 1998: no significant differences in sensitivity or specificity between RFFIT and FAVN

Sources: Smith, Yager & Baer 1973 · Cliquet, Aubert & Sagné 1998 · Briggs et al. 1998 · Wasniewski et al. 2019 (ANSES-EURL) · WOAAH Terrestrial Manual 2024
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The 0.5 IU/mL threshold · Operational correlate of rabies protection

Regulatory decision criterion — not a binary guarantee of individual sterilizing immunity



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